

SemEval-2024 Task 8: Multidomain, Multimodel and Multilingual Machine-Generated Text Detection

Yuxia Wang,[†] Jonibek Mansurov,[†] Petar Ivanov,[†] Jinyan Su,[†] Artem Shelmanov,[†]
Akim Tsvigun,[†] Osama Mohammed Afzal,[†] Tarek Mahmoud,[†]
Giovanni Puccetti,[§] Thomas Arnold,[¶] Chenxi Whitehouse,^{*}
Alham Fikri Aji,[†] Nizar Habash,^{†‡} Iryna Gurevych,[†] Preslav Nakov[†]

[†]MBZUAI, UAE [¶]TU Darmstadt, Germany ^{*}University of Cambridge, UK

[§]Institute of Information Science and Technology, Italy [‡]New York University Abu Dhabi, UAE
{yuxia.wang, jonibek.mansurov, preslav.nakov}@mbzuai.ac.ae

Abstract

We present the results and the main findings of SemEval-2024 Task 8: Multigenerator, Multidomain, and Multilingual Machine-Generated Text Detection. The task featured three subtasks. Subtask A is a binary classification task determining whether a text is written by a human or generated by a machine. This subtask has two tracks: a monolingual track focused solely on English texts and a multilingual track. Subtask B is to detect the exact source of a text, discerning whether it is written by a human or generated by a specific LLM. Subtask C aims to identify the changing point within a text, at which the authorship transitions from human to machine. The task attracted a large number of participants: subtask A monolingual (126), subtask A multilingual (59), subtask B (70), and subtask C (30). In this paper, we present the task, analyze the results, and discuss the system submissions and the methods they used. For all subtasks, the best systems used LLMs.

1 Introduction

The proliferation of Large Language Models (LLMs) has led to a significant increase in the volume of machine-generated text (MGT) across a wide range of domains. This rise has sparked concerns regarding the potential for misuse in fields such as journalism, education, academia, etc (Uchendu et al., 2023; Crothers et al., 2023). Moreover, it poses challenges to maintaining information integrity and ensuring accurate information dissemination. As such, the ability to accurately distinguish between human-written content and machine-generated content has become paramount for identifying potential misuse (Jawahar et al., 2020; Stiff and Johansson, 2022; Macko et al., 2023).

In response to these challenges, we are introducing a shared task that focuses on the detection of machine-generated text across multiple generators,

domains, and languages. We are providing large-scale evaluation datasets for **three subtasks** with the primary goals of fostering extensive research in MGT detection, advancing the development of automated systems for detecting MGT, and reducing instances of misuse:

Subtask A: Human vs. Machine Classification.

The goal of this subtask is to accurately classify a text as either produced by a human or generated by a machine. This is the basic, but one of the most common use-cases of MGT detection systems for preventing the misuse of LLMs. This task is divided into two tracks: (i) The *monolingual track*, which focuses solely on English texts; and (ii) The *multilingual track*, which involves texts in a variety of languages, thereby expanding the diversity and complexity beyond existing benchmarks.

Subtask B: Multi-Way Generator Detection.

This task aims to pinpoint the exact source of a text, i.e., determine whether it originated from a human or a specific LLM (GPT-3, GPT-3.5, GPT-4, Cohere, DALL-E, or BLOOMz). Determining a particular LLM that potentially generated the given text is important from several perspectives: it can help to narrow down the set of LLMs for more sensitive white-box detection techniques or in cases where the generated material is harmful, misleading, or illegal, it might be useful for addressing ethical concerns and legal obligations.

Subtask C: Changing Point Detection.

The goal of this subtask is to precisely identify the exact boundary (changing point) within a text at which the authorship transitions from a human to machine happens. The texts begin with human-written content, which at some point is automatically continued by LLMs (GPT and LLaMA series). The percentage of the human-written section varies from 0% to 50%. This task takes into account the fact

that in many malignant use-cases of LLMs, the part of the text might be written by a human and a part might be generated by a machine. It is hard to classify a text as machine-generated if a big chunk is actually human-written. This is a way to obscure the usage of LLM, and the formulation of Subtask C addresses this challenge.

The task attracted a large number of participants: 126 teams for the Subtask A monolingual track, 59 teams for the Subtask A multilingual track, 70 teams for Subtask B, and 30 teams for Subtask C, with a total of 54 participating teams having submitted a system description paper for all subtasks.

Next, we introduce the MGT detection techniques considered in this shared task in §2; §3 describes the corpus and the evaluation metrics; §4 details the organization of the task; §5 provides an overview of the participating systems; and §6 discusses the evaluation results.

2 Background

Detecting machine-generated text is primarily formulated as a binary classification task (Zellers et al., 2019; Gehrmann et al., 2019a; Solaiman et al., 2019; Ippolito et al., 2019), naively distinguishing between human-written and machine-generated text. In general, there are two main approaches: the supervised methods (Wang et al., 2024a,b; Uchendu et al., 2021; Zellers et al., 2019; Zhong et al., 2020; Liu et al., 2022) and the unsupervised ones, such as zero-shot methods (Solaiman et al., 2019; Ippolito et al., 2019; Mitchell et al., 2023; Su et al., 2023; Hans et al., 2024). While supervised approaches yield relatively better results, they are susceptible to overfitting (Mitchell et al., 2023; Su et al., 2023). Meanwhile, unsupervised methods may require unrealistic white-box access to the generator. In the following, we provide background information on each subtask, respectively.

Subtask A: Mono-lingual and Multi-lingual Binary Classification Given the prevalence of the binary classification task, various benchmarks assess model performance in both mono-lingual and multi-lingual settings. HC3 (Guo et al., 2023) compares ChatGPT-generated text with human-written text in English and Chinese, utilizing logistic regression models trained on GLTR Test-2 features (Gehrmann et al., 2019a) and RoBERTa (Liu et al., 2019)-based classifiers for detection. Benchmark results by Wang et al. (2024b) include

evaluations of several supervised detectors, such as RoBERTa (Liu et al., 2019), XLM-R (Conneau et al., 2019), logistic regression classifier with GLTR features (Gehrmann et al., 2019b), and stylistic features (e.g., stylometry (Li et al., 2014), NELA (Horne et al., 2019) features). Macko et al. (2023) create a similar resource called MULTI-TuDE for 11 languages in the news domain and conduct an extensive evaluation of various baselines. Our effort extends the previous works by providing evaluation setup for multiple domains, multiple languages, and for state-of-the-art LLMs, including ChatGPT and GPT-4.

Subtask B: Multi-Way Generator Detection

Multi-way generator detection, attributing texts not just to their machine-generated nature but also to specific generators, resembles authorship attribution. Munir et al. (2021) find that texts from language models (LMs) have distinguishable features for source attribution. Uchendu et al. (2020) addresses three authorship attribution problems: (1) determining if two texts share the same origin, (2) discerning whether a text is machine or human-generated, and (3) identifying the language model responsible for text generation. Approaches like GPT-who by Venkatraman et al. (2023) employ UID-based features to capture unique signatures of each language model and human author, while Rivera Soto et al. (2024) leverages representations of writing styles.

Subtask C: Change Point Detection Change point detection, which is closely tied to authorship obfuscation (Macko et al., 2024), extends beyond binary/multi-class classification to an adversarial co-authorship setting involving both humans and machines (Dugan et al., 2023). Machine-generated text detection methods are vulnerable to authorship obfuscation attacks such as paraphrasing (Crothers et al., 2022; Krishna et al., 2023; Shi et al., 2023; Koike et al., 2023), back-translation, and change point detection. Related to Subtask C, (Gao et al., 2024) introduces a dataset with mixed machine and human-written texts using operations such as polish, complete (Xie et al., 2023), rewrite (Shu et al., 2023), humanize (adding natural noise (Wang et al., 2021)), and adapt (Gero et al., 2022). Kumarage et al. (2023) uses stylometric signals to quantify changes in tweets and detect when AI starts generating tweets. Different to our task, they focus on human-to-AI author changes within a given Twitter

Split	Source	davinci-003	ChatGPT	Cohere	Dolly-v2	BLOOMz	GPT-4	Machine	Human
Train	Wikipedia	3,000	2,995	2,336	2,702	–	–	11,033	14,497
	WikiHow	3,000	3,000	3,000	3,000	–	–	12,000	15,499
	Reddit	3,000	3,000	3,000	3,000	–	–	12,000	15,500
	arXiv	2,999	3,000	3,000	3,000	–	–	11,999	15,498
	PeerRead	2,344	2,344	2,342	2,344	–	–	9,374	2,357
Dev	Wikipedia	–	–	–	–	500	–	500	500
	WikiHow	–	–	–	–	500	–	500	500
	Reddit	–	–	–	–	500	–	500	500
	arXiv	–	–	–	–	500	–	500	500
	PeerRead	–	–	–	–	500	–	500	500
Test	Outfox	3,000	3,000	3,000	3,000	3,000	3,000	18,000	16,272

Table 1: **Subtasks A: Monolingual Binary Classification.** Data statistics over Train/Dev/Test splits

Split	Language	davinci-003	ChatGPT	LLaMA2	Jais	Other	Machine	Human
Train	English	11,999	11,995	–	–	35,036	59,030	62,994
	Chinese	2,964	2,970	–	–	–	5,934	6,000
	Urdu	–	2,899	–	–	–	2,899	3,000
	Bulgarian	3,000	3,000	–	–	–	6,000	6,000
	Indonesian	–	3,000	–	–	–	3,000	3,000
Dev	Russian	500	500	–	–	–	1,000	1,000
	Arabic	–	500	–	–	–	500	500
	German	–	500	–	–	–	500	500
Test	English	3,000	3,000	–	–	9,000	15,000	13,200
	Arabic	–	1,000	–	100	–	1,100	1,000
	German	–	3,000	–	–	–	3,000	3,000
	Italian	–	–	3,000	–	–	3,000	3,000

Table 2: **Subtasks A: Multilingual Binary Classification.** Data statistics over Train/Dev/Test splits (Others generators are Cohere, Dolly-v2 and BLOOMz)

timeline.

3 Dataset and Metrics

In this section, we describe the datasets and evaluation metrics for all subtask tracks, including the size, domains, generators, and language distribution across training, development, and test splits.

3.1 Subtask A: Monolingual Track

Data: Table 1 presents statistics across generators, domains, and splits. The training set encompasses domains such as Wikipedia, WikiHow, Reddit, arXiv, and PeerRead, comprising a total of 56,400 machine-generated and 63,351 human-written texts. BLOOMz is utilized as an unseen generator in the development set, which contains 2,500 machine-generated and 2,500 human-written texts. For the test set, OUTFOX is introduced as the surprising domain, and GPT-4 serves as the surprising generator, with a dataset of 18,000 machine-generated and 16,272 human-written texts.

Metrics: Accuracy is used to evaluate detectors.

3.2 Subtask A: Multilingual Track

Data: Table 2 presents the dataset statistics. The training set encompasses texts in English, Chinese, Urdu, Bulgarian, and Indonesian, totaling 76,863 machine-generated and 80,994 human-written texts. The development set includes Arabic (sourced

Split	Source	davinci-003	ChatGPT	Cohere	Dolly-v2	BLOOMz	Human
Train	Wikipedia	3,000	2,995	2,336	2,702	2,999	3,000
	WikiHow	3,000	3,000	3,000	3,000	3,000	2,995
	Reddit	3,000	3,000	3,000	3,000	2,999	3,000
	arXiv	2,999	3,000	3,000	3,000	3,000	2,998
Dev	PeerRead	500	500	500	500	500	500
Test	Outfox	3,000	3,000	3,000	3,000	3,000	3,000

Table 3: **Subtasks B: Multi-Way Generator Detection.** Data statistics over Train/Dev/Test splits

Domain	Generator	Train	Dev	Test	Total
PeerRead	ChatGPT	3,649 (232)	505 (23)	1,522 (89)	5,676 (344)
	LLaMA-2-7B*	3,649 (5)	505 (0)	1,035 (1)	5,189 (6)
	LLaMA-2-7B	3,649 (227)	505 (24)	1,522 (67)	5,676 (318)
	LLaMA-2-13B	3,649 (192)	505 (24)	1,522 (84)	5,676 (300)
	LLaMA-2-70B	3,649 (240)	505 (21)	1,522 (88)	5,676 (349)
OUTFOX	GPT-4	–	–	1,000 (10)	1,000 (10)
	LLaMA2-7B	–	–	1,000 (8)	1,000 (8)
	LLaMA2-13B	–	–	1,000 (5)	1,000 (5)
	LLaMA2-70B	–	–	1,000 (19)	1,000 (19)
Total	all	18,245	2,525	11,123	31,893

Table 4: **Subtask C: Change Point Detection.** We use generators GPT and LLaMA-2 series over domains of academic paper review (PeerRead) and student essay (OUTFOX). The number in “()” is the number of examples purely generated by LLMs, i.e., human and machine boundary index=0. LLaMA-2-7B* and LLaMA-2-7B used different prompts. Bold data is used in shared task training, development, and test.

from Wikipedia), Russian, and German (sourced from Wikipedia), each contributing 2,000 texts from both machine-generated and human-written sources. In the test set, Italian is introduced as the unexpected language, with *OUTFOX* and *News* serving as new domains for English, Arabic, and German texts. This set comprises 22,100 machine-generated and 20,200 human-written texts.

Metrics: Accuracy is used to evaluate detectors.

3.3 Subtask B

Data: In Table 3, we incorporate texts from five generators (davinci-003, ChatGPT, Cohere, Dolly-v2, and BLOOMz) alongside human-written texts. The development set features texts from the PeerRead domain, while the test set introduces OUTFOX (specifically, student essays) as the unexpected domain.

Metrics: Accuracy is used to evaluate detectors.

3.4 Subtask C

Data: The training and development sets for subtask C are PeerRead ChatGPT generations, with **5,349** and **505** examples respectively (first row of Table 4), and the test set is the combination of the *test column* of Table 4, totaling 11,123 examples.